SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ

COURSE NAME: COMPUTER VISION **COURSE CODE:ITA0515** **Slot B**

COURSE OUTCOME COVERED
CO3: ***Analyze and extract image features using detection and matching techniques for effective image representation and comparison.*** **(BTL 4)**
Assessment Tool Weightage: 10%

QUESTIONS:50 **Total Marks 50** **Duration :60 Minutes**
Annexure A
MCQ

Code of Conduct:

I S.Jeswanth kumar-192321056 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Jeswanth kumar

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	*Marks(100)*
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Image Normalization (Preprocessing)

An image pixel value **r = 150** is normalized to the range **[0,1]** from **8-bit** data.

What is the normalized value?

A) 0.39
B) 0.49
C) 0.59
D) 0.78

Answer: C) 0.59

2. Image Storage Calculation

A grayscale image of size **1024 × 512** is stored using **8 bits per pixel**.

What is the storage requirement?

A) 256 KB
B) 512 KB
C) 1024 KB
D) 2048 KB

Answer: B) 512 KB

3. Edge Detection – Gradient Magnitude

Given Sobel gradients:
($G_x = 6$), ($G_y = 8$)

What is the gradient magnitude?

A) 10
B) 12
C) 14
D) 16

Answer: A) 10

4. Edge Direction Calculation

For the same gradients ($G_x = 6$), ($G_y = 8$),
what is the **edge direction**?

A) 36.8°
B) 45°
C) 53.1°

Answer: C) 53.1°

5. Harris Corner Detection Response

Given:
(R = \det(M) - k(\text{trace}(M))^2)
Where:
det(M) = **200**, trace(M) = **30**, k = **0.04**

What is the corner response R?

A) 136
B) 164
C) 180
D) 200

Answer: B) 164

6. Harris Corner Classification

If the Harris response **R = +250**, the pixel is classified as:

A) Flat region
B) Edge
C) Corner
D) Noise

Answer: C) Corner

7. Hough Transform – Line Representation

A line detected at **$\theta = 90°$** and **$\rho = 50$** corresponds to:

A) Horizontal line at y = 50
B) Vertical line at x = 50
C) Diagonal line
D) Random line

Answer: B) Vertical line at x = 50

An image is scaled by a factor of **2**.
If original keypoint scale is **$\sigma = 1.6$**, new scale is:

A) 0.8
B) 1.6
C) 3.2
D) 4.0

Answer: C) 3.2

9. PCA Dimensionality Reduction

A feature vector has **100 dimensions**.
PCA retains **95% variance using 20 components**.

What is the dimensionality after PCA?

A) 5
B) 20
C) 80
D) 95

Answer: B) 20

10. PCA Covariance Matrix Size

For a dataset with **50 features**, what is the size of the covariance matrix?

A) 50 × 1
B) 1 × 50
C) 50 × 50
D) 100 × 100

Answer: C) 50 × 50

11. Image Quantization (Preprocessing)

A grayscale image is quantized from **256 levels to 32 levels**.

What is the **number of bits per pixel after quantization**?

A) 3
B) 4
C) 5
D) 6

Answer: C) 5

12. Image Storage Reduction

An RGB image of size **512 × 512** uses **8 bits per channel**.
It is converted to grayscale with **8 bits per pixel**.

What is the **percentage reduction in storage**?

A) 33%
B) 50%
C) 66.7%
D) 75%

Answer: C) 66.7%

13. Sobel Edge Response

Given Sobel kernel response values:
($G_x = -12$), ($G_y = 5$)

What is the **gradient magnitude (approx.)**?

A) 10
B) 12
C) 13
D) 15

Answer: C) 13

14. Gradient Direction Calculation

For ($G_x = -12$), ($G_y = 5$), the gradient direction is closest to:

A) 22°
B) 157°
C) 180°
D) −22°

Answer: B) 157°

15. Harris Corner Detection Response

Given:
det(M) = **400**
trace(M) = **40**
k = **0.06**

Compute the Harris response **R**.

A) 256
B) 304
C) 352
D) 400

Answer: B) 304

16. Harris Corner Decision

For a pixel, eigenvalues of matrix **M** are:
λ_1 = **120**, λ_2 = **130**

This pixel is classified as:

A) Flat region
B) Edge
C) Corner
D) Noise

Answer: C) Corner

17. Hough Transform Voting

A point contributes votes to **180 values of θ** in Hough space.

What is the **shape of its representation**?

A) A point
B) A straight line
C) A sinusoidal curve
D) A circle

Answer: C) A sinusoidal curve

18. Hough Line Detection

A detected line has parameters **$\rho = 0$**, **$\theta = 45°$**.

Which line is represented?

A) Horizontal through origin
B) Vertical through origin
C) Diagonal line through origin
D) Random orientation

Answer: C) Diagonal line through origin

19. SIFT Keypoint Descriptor Size

Each SIFT descriptor consists of **4×4 histograms**, each with **8 bins**.

What is the descriptor dimensionality?

A) 64
B) 96
C) 128
D) 256

Answer: C) 128

20. PCA Eigenvalue Selection

Eigenvalues of covariance matrix:
[40, 25, 15, 10, 5, 5]

How many principal components are needed to retain **80% variance**?

A) 2
B) 3
C) 4
D) 5

Answer: B) 3

21. Preprocessing – Normalization Scenario

A surveillance camera outputs **8-bit grayscale images**.
Before feature extraction, pixel values must be normalized to **[0,1]**.
A pixel has value **204**.

What is the normalized value?

A) 0.60
B) 0.70
C) 0.80
D) 0.90

Answer: C) 0.80

22. Image Representation Scenario

A medical image of size **2048 × 1024** is stored using **16 bits per pixel**.

What is the **memory required**?

A) 2 MB
B) 4 MB
C) 8 MB
D) 16 MB

Answer: B) 4 MB

23. Edge Detection Scenario

At a pixel location, Sobel gradients are:
($G_x = 9$), ($G_y = 12$)

What is the **edge magnitude** at that pixel?

A) 15
B) 18
C) 21
D) 25

Answer: A) 15

24. Edge Orientation Scenario

Using the same gradients ($G_x = 9$), ($G_y = 12$),
what is the **edge direction (approx.)**?

A) 36.8°
B) 45°
C) 53.1°
D) 60°

Answer: C) 53.1°

25. Harris Corner Detection Scenario

For a region in an image, the Harris matrix values are:
det(M) = **300**, trace(M) = **25**, k = **0.04**

What is the **Harris response R**?

A) 200
B) 250
C) 275
D) 300

Answer: B) 275

26. Harris Corner Interpretation Scenario

If the Harris response **R is large and positive**, the pixel corresponds to:

A) Flat region
B) Edge
C) Corner
D) Noise

Answer: C) Corner

2**7. Hough Transform Scenario**

A point in image space votes for all possible **θ values** in Hough space.

What shape does this vote form?

A) A point
B) A straight line
C) A sinusoidal curve
D) A circle

Answer: C) A sinusoidal curve

28. Hough Line Detection Scenario

A detected line has parameters **ρ = 40**, **θ = 90°**.

Which line is represented?

A) Vertical line x = 40
B) Horizontal line y = 40
C) Diagonal line

Answer: B) Horizontal line y = 40

29. SIFT Feature Scenario

An image is scaled by a factor of **0.5**.
A SIFT keypoint detected at scale **σ = 2.0** originally will now appear at:

B) 0.5
A) 1.0
C) 2.0
D) 4.0

Answer: A) 1.0

30. PCA Dimensionality Reduction Scenario

A feature vector has **200 dimensions**.
After PCA, **30 principal components** retain **92% variance**.

What is the **new feature dimension**?

A) 30
B) 92
C) 170
D) 200

Answer: A) 30

31. Preprocessing – Bit Depth Conversion

A grayscale image originally stored in **12-bit format** is converted to **8-bit** before processing.
An original pixel value is **3000**.

What is the approximate 8-bit value after linear scaling?

A) 120
B) 150
C) 187
D) 225

Answer: C) 187

32. Image Representation – Sampling Scenario

A camera samples an image at **600 × 400** pixels.
The image is down-sampled by a factor of **2 in both directions**.

What is the new image size?

A) 150 × 100
B) 300 × 200
C) 400 × 300
D) 600 × 400

Answer: B) 300 × 200

33. Edge Detection – Sobel Response

At a pixel, Sobel gradients are:
(G_x = -8), (G_y = 15)

What is the **edge magnitude (approx.)**?

A) 13
B) 15
C) 17
D) 20

Answer: C) 17

34. Edge Orientation Scenario

For (G_x = -8), (G_y = 15), the gradient direction is closest to:

A) 28°
B) 62°
C) 118°
D) 152°

Answer: C) 118°

35. Harris Corner Detection – Computation

Given:
det(M) = **500**, trace(M) = **35**, k = **0.05**

What is the Harris response **R**?

A) 439
B) 450
C) 469

Answer: A) 439

36. Harris Corner Classification

If one eigenvalue is large and the other is small, the pixel represents:

A) Flat region
B) Edge
C) Corner
D) Noise

Answer: B) Edge

37. Hough Transform – Line Interpretation

A line detected with parameters **ρ = 0**, **θ = 90°** corresponds to:

A) Vertical line at x = 0
B) Horizontal line at y = 0
C) Diagonal line

Answer: B) Horizontal line at y = 0

38. SIFT Descriptor Scenario

Each SIFT descriptor contains **4 × 4** blocks with **8 orientation bins** each.

How many values describe one keypoint?

A) 32
B) 64
C) 96

Answer: C) 128

39. PCA Variance Retention

Eigenvalues of covariance matrix are:
[50,; 30,; 10,; 5,; 5]

How many principal components are needed to retain **90% variance**?

A) 1
B) 2
C) 3
D) 4

Answer: C) 3

40. PCA Projection Scenario

Original feature vector dimension = **120**.
PCA reduces it to **25 components**.

What is the **reduction percentage**?

A) 60%
B) 70%
C) 79%

Answer: C) 79%

41. Quantization Error Analysis (Preprocessing)

An 8-bit grayscale image is quantized to **64 levels**.
Original pixel value = **150**.

What is the **maximum possible quantization error**?

A) ±1
B) ±2
C) ±4
D) ±8

Answer: B) ±2

42. Image Representation Storage Comparison

Image A: **1024 × 1024 grayscale (8-bit)**
Image B: **1024 × 1024 RGB (8-bit per channel)**

What is the **storage ratio (B : A)**?

A) 1 : 1
B) 2 : 1
C) 3 : 1
D) 4 : 1

Answer: C) 3 : 1

43. Sobel Edge Strength Comparison

At two pixels:

Pixel-1: (G_x = 6,; G_y = 8)
Pixel-2: (G_x = 9,; G_y = 12)

Which pixel corresponds to a **stronger edge**?

A) Pixel-1
B) Pixel-2
C) Both equal

Answer: B) Pixel-2

44. Harris Corner Response Comparison

Two regions have the following Harris parameters:

Region	det(M)	trace(M)
R1	300	25
R2	350	40

Given **k = 0.04**, which region is **more likely a corner**?

A) R1
B) R2
C) Both
D) Neither

Answer: B) R2

45. Harris Corner vs Edge Decision

Eigenvalues of matrix **M** at a pixel are:
λ_1 = **120**, λ_2 = **10**

What does this pixel represent?

A) Flat region
B) Edge
C) Corner
D) Noise

Answer: B) Edge

46. Hough Transform Voting Analysis

Two points generate curves in Hough space that intersect at **(ρ = 60, θ = 45°)**.

What does this intersection indicate?

A) A corner
B) A circle
C) A line in image space
D) Noise accumulation

Answer: C) A line in image space

47. Hough Transform Parameter Effect

Increasing the **θ resolution** from 1° to 0.5° results in:

A) Fewer accumulator cells
B) Lower memory usage
C) Higher line detection accuracy

Answer: C) Higher line detection accuracy

48. SIFT Scale Analysis

An object appears at scale **σ = 1.6** in Image-1.
In Image-2, the object is scaled by **1.5×**.

At what scale should the corresponding SIFT keypoint appear?

A) 1.1
B) 1.6
C) 2.4

Answer: C) 2.4

49. PCA Variance Contribution Analysis

Eigenvalues of covariance matrix are:
[60,; 20,; 10,; 5,; 5]

How many principal components are needed to retain **90% variance**?

A) 1
B) 2
C) 3
D) 4

Answer: C) 3

50. PCA Dimensionality Reduction Impact

Original feature dimension = **150**
After PCA = **30 components**

What is the **percentage reduction**?

A) 60%
B) 70%
C) 80%
D) 85%

Answer: C) 80%